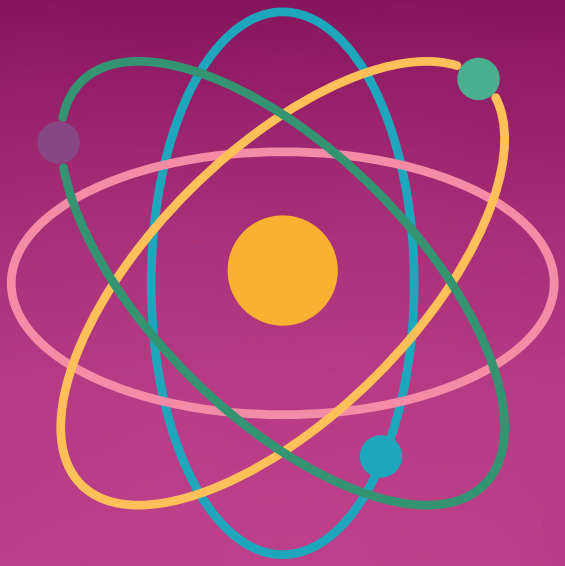
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PHYSICAL CHEMISTRY

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Ionic Equilibrium

English Medium

EXERCISE # S-I

IONIZATION CONSTANTS AND pH

Q.1 Calculate the number of H^+ present in one ml of solution whose pH is 13.

IE0001

Q.2 (i) K_w for H_2O is 9×10^{-14} at 60°C. What is pH of water at 60°C. (log 3 = 0.47)

IE0002

(ii) What is the nature of solution at 60°C whose **IE0003**

(a) pH = 6.7 (b) pH = 6.35

Q.3 The value of K_w at the physiological temperature (37°C) is 2.56×10^{-14}. What is the pH at the neutral point of water at this temperature? (log 2 = 0.3)

IE0004

Q.4 Calculate pH of following solutions :

(a) 0.1 M HCl **IE0005**

(b) 0.1 M CH_3COOH ($K_a = 1.8 \times 10^{-5}$) (log $\sqrt{1.8} = 0.13$) **IE0006**

(c) 0.1 M NH_4OH ($K_b = 1.8 \times 10^{-5}$) **IE0007**

(d) 10^{-8} M HCl [$\sqrt{401} = (20.02)$] [log 1.051 = 0.03] **IE0008**

(e) 10^{-10} M NaOH **IE0009**

(f) 10^{-6} M CH_3COOH ($K_a = 1.8 \times 10^{-5}$) **IE0010**

(g) 10^{-8} M CH_3COOH ($K_a = 1.8 \times 10^{-5}$) [$\sqrt{401} = (20.02)$] [log 1.051 = 0.03] **IE0011**

(h) Decimolar solution of Baryta ($Ba(OH)_2$), diluted 100 times. (log2 = 0.3) **IE0012**

(i) 10^{-3} mole of KOH dissolved in 100 L of water. **IE0013**

(j) Equal volume of HCl solution (PH = 4) + 0.0019 N HCl solution **IE0014**

Q.5 Calculate :

(a) K_a for a monobasic acid whose 0.10 M solution has pH of 4.50.

(b) K_b for a monoacidic base whose 0.10 M solution has a pH of 10.50.

IE0015

Q.6 Calculate the ratio of degree of dissociation (α_2/α_1) when an acetic acid solution is diluted 100 times. Assume $\alpha << 1$, even on dilution. [Given $K_a = 10^{-5}$ M]

IE0016

Q.7 Calculate the ratio of degree of dissociation of acetic acid and hydrocyanic acid (HCN) in 1 M their respective solution of acids.[Given $K_{a(CH_3COOH)} = 1.8 \times 10^{-5}$; $K_{a(HCN)} = 6 \times 10^{-10}$]

IE0017

Q.8 How many moles of HCl must be removed from 1 litre of aqueous HCl solution to change its pH from 2 to 3 ?

IE0018

Q.9 pH of a dilute solution of HCl is 6.95. Calculate molarity of HCl solution.

$$\begin{bmatrix} 10^{-6.95} = 11.22 \times 10^{-8} \\ 10^{-7.05} = 8.90 \times 10^{-8} \end{bmatrix}$$

IE0205

Q.10 The pH of aqueous solution of ammonia is 10. Find molarity of solution. K_b (NH_4OH) = 10^{-5}.

IE0019

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Q.11 The solution of weak monoprotic acid which is 0.01 M, has pH = 3. Calculate K_a of weak acid.

IE0020

Q.12 Boric acid is a weak monobasic acid. It ionizes in water as

$B(OH)_3 + H_2O \rightleftharpoons B(OH)_4^- + H^+ : K_a = 8 \times 10^{-10}$

Calculate pH of 0.5 M boric acid.

IE0021

MIXTURE OF TWO OR MORE ACIDS / BASES

Q.13 The pH of the solution produced when an aqueous solution of strong acid pH 5 is mixed with equal volume of an aqueous solution of strong acid of pH 3 is :-

IE0022

Q.14 Calculate pH of following solutions : [log 0.3 = – 0.522]

(a) 0.1 M H_2SO_4 (50 ml) + 0.4 M HCl 50 (ml)

(b) 0.1 M HA + 0.1 M HB [K_a (HA) = 5×10^{-5} ; K_a (HB) = 4×10^{-5}]

IE0023

Q.15 Calculate pH of a solution containing 0.1M HA (Ka = 10^{-5}) & 0.1 M HCl.

IE0024

Q.16 Calculate $[H^+]$ and $[CHCl_2COO^-]$ in a solution that is 0.01 M in HCl and 0.01 M in $CHCl_2COOH$.

Take ($K_a = 3 \times 10^{-2}$) $\left(\sqrt{28} = 5.3\right)$

IE0206

Q.17 Calculate $[H^+]$, $[CH_3COO^-]$ and $[C_7H_5O_2^-]$ in a solution that is 0.2 M in acetic acid and 0.1M in benzoic acid. K_a(acetic) = 1.8×10^{-5} , K_a (benzoic) = 5.4×10^{-5}.

IE0207

POLYPROTIC ACIDS & BASES

Q.18 What are the concentration of H^+, $H_2C_2O_4$, $HC_2O_4^-$ and $C_2O_4^{2-}$ in a 0.1 M solution of oxalic acid ?

[$K_1 = 10^{-2}$ M and $K_2 = 10^{-5}$ M] $\left[\sqrt{41} = 6.4\right]$

IE0025

Q.19 Calculate $[H^+]$, $[H_2PO_4^-]$, $[HPO_4^{2-}]$ and $[PO_4^{3-}]$ in a 0.01M solution of H_3PO_4.

Take $K_1 = 10^{-3}$, $K_2 = 10^{-8}$, $K_3 = 10^{-13}$, $\sqrt{41} = 6.4$

IE0208

Q.20 Calculate pH of 0.2 M – $B(OH)_2$ solution.

($K_{b_1} = 2 \times 10^{-5}$; $K_{b_2} = 4 \times 10^{-11}$, log2 = 0.3)

IE0026

HYDROLYSIS

Q.21 What is the OH^- concentration of a 0.18 M solution of CH_3COONa. [$K_a(CH_3COOH) = 1.8 \times 10^{-5}$]

IE0027

Q.22 Calculate the pH of a 2.0 M solution of NH_4Cl. [K_b (NH_3) = 2×10^{-5}]

IE0028

Q.23 0.25 M solution of pyridinium chloride $C_5H_6N^+Cl^-$ was found to have a pH of 2.699. What is K_b for pyridine, C_5H_5N ? (log2 = 0.3010)

IE0029

Q.24 Calculate the extent of hydrolysis & the pH of 0.02 M CH_3COONH_4.

[K_b (NH_3)= 1.8×10^{-5}, $K_a(CH_3COOH) = 1.8 \times 10^{-5}$]

IE0030

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Q.25 Calculate the percent hydrolysis in a 0.06 M solution of KCN. [K_a(HCN) = 6×10^{-10}]

IE0031

Q.26 Calculate the extent of hydrolysis of 0.005 M K_2CrO_4. [$K_2 = 3.2 \times 10^{-7}$ for H_2CrO_4] (It is essentially strong for first ionization).

IE0209

Q.27 A 0.010 M solution of $PuO_2(NO_3)_2$ was found to have a pH of 4.0. What is the hydrolysis constant, K_h, for PuO_2^{2+} ,and what is K_b for PuO_2OH^+ ?

IE0210

Q.28 What is the pH of 0.1M $NaHCO_3$? $K_1 = 5 \times 10^{-7}$, $K_2 = 5 \times 10^{-11}$ for carbonic acids.

IE0032

Q.29 Calculate pH of 0.05M potassium hydrogen phthalate, $KHC_8H_4O_4$.

$$H_2C_8H_4O_4 + H_2O \rightleftharpoons H_3O^+ + HC_8H_4O_4^- \qquad pK_1 = 2.94$$

$$HC_8H_4O_4^- + H_2O \rightleftharpoons H_3O^+ + C_8H_4O_4^{2-} \qquad pK_2 = 5.44$$

IE0211

Q.30 The acid ionization (hydrolysis) constant of Zn^{2+} is 1.0×10^{-9}
(a) Calculate the pH of a 0.001 M solution of $ZnCl_2$
(b) What is the basic dissociation constant of $Zn(OH)^+$?

IE0033

BUFFER SOLUTION

Q.31 Calculate the pH of solution containing 0.1M - HCN and 0.1M - NaCN. Ka of HCN = 10^{-9}

IE0034

Q.32 Calculate the pH of solution containing 0.2 M - NH_4OH and 0.1 M – NH_4Cl. K_b of NH_4OH = 1.8×10^{-5}. (log2 = 0.3, log 1.8 = 0.26)

IE0035

Q.33 0.4 mole CH_3COONa is added in 500 ml 0.4 M –CH_3COOH solutions. What is the pH of final solution ? K_a of $CH_3COOH = 1.8 \times 10^{-5}$. (log2 = 0.3, log 1.8 = 0.26).

IE0036

Q.34 A buffer of pH 9.26 is made by dissolving x moles of ammonium sulphate and 0.1 mole of ammonia into 100 mL solution. If pK_b of ammonia is 4.74, calculate value of x.

IE0037

Q.35 Determine [OH^-] of a 0.050 M solution of ammonia to which sufficient NH_4Cl has been added to make the total [NH_4^+] equal to 0.100.[$K_{b(NH_3)} = 1.8 \times 10^{-5}$, $pK_b = 4.74$]

IE0038

Q.36 Calculate the pH of a solution containing 0.2 M HCO_3^- and 0.1 M CO_3^{2-}
[$K_1(H_2CO_3) = 4 \times 10^{-7}$; $K_2(HCO_3^-) = 4 \times 10^{-11}$]

IE0039

Q.37 Calculate the pH of a solution prepared by mixing 50.0 mL of 0.200 M $HC_2H_3O_2$ and 50.0 mL of 0.100 M NaOH.[$K_{a(CH_3COOH)} = 1.8 \times 10^{-5}$, $pK_a = 4.74$]

IE0040

Q.38 50 mL of 0.1 M NaOH is added to 75 mL of 0.1 M NH_4Cl to make a basic buffer. If pK_a of NH_4^+ is 9.26, calculate pH.

IE0041

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Q.39 Calculate the pH of a solution which results from the mixing of 50.0 ml of 0.3 M HCl with 50.0 ml of 0.4 M NH_3 . [K_b (NH_3) = 1.8×10^{-5}, $pK_b = 4.74$]

IE0042

Q.40 In 100 ml buffer solution of 0.1M CH_3COOH & 0.1M CH_3COONa, how many millimoles of NaOH should be added to increase it's pH by 0.3.
Given (log 2 = 0.3)

IE0043

ACID BASE REACTIONS & TITRATIONS

Q.41 Calculate OH^- concentration at the equivalent point when a solution of 0.2 M acetic acid is titrated with a solution of 0.2 M NaOH. K_a for the acid = 10^{-5}.

IE0044

Q.42 Calculate the hydronium ion concentration and pH at the equivalence point in the reaction of 22.0 mL of 0.10M acetic acid, CH_3COOH, with 22.0 mL of 0.10 M NaOH. [$K_a = 2 \times 10^{-5}$]

IE0045

Q.43 Calculate the hydronium ion concentration and the pH at the equivalence point in a titration of 50.0 mL of 0.40 M NH_3 with 0.40M HCl.[$K_b = 2 \times 10^{-5}$]

IE0046

Q.44 CH_3COOH (50 ml, 0.1 M) is titrated against 0.1 M NaOH solution. Calculate the pH at the addition of 0 ml, 10 ml 20 ml, 25 ml, 40 ml, 50 ml of NaOH. K_a of CH_3COOH is 2×10^{-5}.
[log 2 = 0.3010, log3 = 0.4771]

IE0047

INDICATORS

Q.45 For the acid indicator thymol blue, pH is 3 when half the indicator is in unionised form. Find the % of indicator in unionised form in the solution with $[H^+] = 4 \times 10^{-3}$ M.

IE0048

Q.46 Bromophenol blue is an acid indicator with a K_a value of 6×10^{-5} . What % of this indicator is in its basic form at a pH of 5 ?

IE0049

Q.47 At what pH does an indicator change colour if the indicator is a weak acid with $K_{ind} = 4 \times 10^{-4}$. For which one(s) of the following neutralizations would the indicator be useful ? Explain.
(a) NaOH + CH_3COOH (b) HCl + NH_3 (c) HCl + NaOH

IE0050

Q.48 An acid indicator has a K_a of 3×10^{-5}. The acid form of the indicator is red & the basic form is blue. By how much must the pH change in order to change the indicator form 75% red to 75 % blue? [log 3 = 0.4771]

IE0051

SOLUBILITY & SOLUBILITY PRODUCT'S

Q.49 The values of K_{sp} for the slightly soluble salts MX and QX_2 are each equal to 4.0×10^{-18}. Which salt is more soluble? Explain your answer fully.

IE0052

Q.50 The solubility of $PbSO_4$ in water is 0.0608 g/L. Calculate the solubility product constant of $PbSO_4$. Molar mass $PbSO_4$ = 304 g/mole

IE0053

Q.51 How many mole CuI ($K_{sp} = 5 \times 10^{-12}$) will dissolve in 1.0 L of 0.10 M NaI solution ?

IE0054

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Q.52 A solution of saturated CaF_2 is found to contain 4×10^{-4} M fluoride ion. Calculate the K_{sp} of CaF_2. Neglect hydrolysis.

IE0055

Q.53 The solubility of ML_2 (formula weight = 60 g/mol) in water is 2.4×10^{-5} g/100 mL solution. Calculate the solubility product constant for ML_2.

IE0056

Q.54 Calculate the solubility of A_2X_3 in pure water, assuming that neither kind of ion reacts with water. For A_2X_3 , $K_{sp} = 1.08 \times 10^{-23}$

IE0057

Q.55 Determine the solubility of AgCl in 0.1 M $BaCl_2$. [K_{sp} for AgCl = 1×10^{-10}]

IE0058

Q.56 Calculate solubility of $Ca_3(PO_4)_2$ ($K_{sp} = 10^{-15}$) in presence of 0.1 M $CaCl_2$ solution.

IE0059

SIMULTANEOUS SOLUBILITY

Q.57 Calculate the Simultaneous solubility of AgSCN and AgBr. K_{sp} (AgSCN) = 3.2×10^{-12}, K_{sp}(AgBr) = 8×10^{-13}.

IE0060

Q.58 Calculate F^- in a solution saturated with respect of both MgF_2 and SrF_2. $K_{sp}(MgF_2) = 9.5 \times 10^{-9}$, $K_{sp}(SrF_2) = 4 \times 10^{-9}$.

IE0212

COMPLEX FORMATION

Q.59 Calculate the solubility of AgCl in 0.2 M - NH_3 solution.
Given : K_{sp} of AgCl = 2×10^{-10} , K_f of $Ag(NH_3)_2^+ = 8 \times 10^6$.

IE0061

Q.60 Calculate the solubility of AgCN in 0.4 M - KCN solution
(i) neglecting complex formation
(ii) considering complex formation. Given : Ksp of AgCN = 8×10^{-10}, K_d of $Ag(CN)_2^-$ = 4×10^{-8}.

IE0213

SOLUBILITY, CONSIDERING HYDROLYSIS

Q.61 Calculating the solubility of MX in water. Also calculate pH of solution.
Given : K_{sp} of MX = 4×10^{-8} ; K_a of HX = 2×10^{-6} and MOH is strong base.

IE0214

Q.62 Calculate the solubility of AgCN in a buffer solution at pH = 3.0. .
Given : K_{sp} of AgCN = 8×10^{-10} , K_a of HCN = 5×10^{-10}.

IE0062

PRECIPITATION

Q.63 A solution has a Mg^{2+} concentration of 0.0010 mol/L. Will $Mg(OH)_2$ precipitate if the OH^- concentration of the solution is [$K_{sp} = 1.2 \times 10^{-11}$]
(a) 10^{-5} mol/L (b) 10^{-3} mol/L ?

IE0063

Q.64 200 ml of 2×10^{-4}M – $AgNO_3$ solution is mixed with 400 ml of 1.2×10^{-6} M – NaCl solution. Predict whether precipitation of AgCl will occur or not. K_{sp} of AgCl = 2×10^{-10}.

IE0064

Q.65 Calculate the minimum mass of Na_2SO_4 needed to just start precipitation of $BaSO_4$ from 500 ml of 2×10^{-5}M - $BaCl_2$ solution. K_{sp} of $BaSO_4$ = 8×10^{-8}.

IE0065

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EXERCISE # S-II

Q.1 What are the concentrations of H^+, HSO_4^-, SO_4^{2-} and H_2SO_4 in a 0.20 M solution of sulphuric acid?

Given: $H_2SO_4 \longrightarrow H^+ + HSO_4^-$; strong

$HSO_4^- \rightleftharpoons H^+ + SO_4^{2-}$; $K_2 = 10^{-2}$ M

IE0066

Q.2 Calculate the pH of a 0.1M solution of $H_2NCH_2CH_2NH_2$; ethylenediamine (en). Determine the en H_2^{2+} concentration in the solution. K_{b_1} and K_{b_2} values of ethylenediamine are 9×10^{-5} and 7.1×10^{-8} respectively.

IE0215

Q.3 Nicotine, $C_{10}H_{14}N_2$, has two basic nitrogen atoms and both can react with water to give a basic solution

Nic (aq) + H_2O (l) $\rightleftharpoons$ $NicH^+$ (aq) + OH^- (aq)

$NicH^+$ (aq) + H_2O (l) $\rightleftharpoons$ $NicH_2^{2+}$ (aq) + OH^- (aq)

K_{b_1} is 8×10^{-7} and K_{b_2} is 10^{-10}. Calculate the approximate pH of a 0.20 M solution.

IE0067

Q.4 Determine the $[S^{2-}]$ in a saturated (0.1M) H_2S solution to which enough HCl has been added to produce a $[H^+]$ of 2×10^{-4} . $K_1 = 10^{-7}$, $K_2 = 10^{-14}$.

IE0216

Q.5 An aqueous solution contains 0.01 M RNH_2 ($K_b = 2 \times 10^{-6}$) & 10^{-4} M NaOH. The concentration of OH^- is nearly :

IE0068

Q.6 Calculate the pH of 1.0×10^{-3} M sodium phenoxide, $NaOC_6H_5$. K_a for HOC_6H_5 is 0.6×10^{-10}.

IE0069

Q.7 Calculate the OH^- concentration and the H_3PO_4 concentration of a solution prepared by dissolving 0.1 mol of Na_3PO_4 in sufficient water to make 1L of solution. $K_1 = 7.1 \times 10^{-3}$, $K_2 = 6.3 \times 10^{-8}$, $K_3 = 4.5 \times 10^{-13}$.

IE0217

Q.8 Calculate the pH of 0.1 M solution of (i) $NaHCO_3$, (ii) Na_2HPO_4 and (iii) NaH_2PO_4. Given that:

$CO_2 + H_2O \rightleftharpoons H^+ + HCO_3^-$;	$K_1 = 4.2 \times 10^{-7}$ M
$HCO_3^- \rightleftharpoons H^+ + CO_3^{2-}$;	$K_2 = 4.8 \times 10^{-11}$ M
$H_3PO_4 \rightleftharpoons H^+ + H_2PO_4^-$;	$K_1 = 7.5 \times 10^{-3}$ M
$H_2PO_4^- \rightleftharpoons H^+ + HPO_4^{2-}$;	$K_2 = 6.2 \times 10^{-8}$ M
$HPO_4^{2-} \rightleftharpoons H^+ + PO_4^{3-}$;	$K_3 = 1.0 \times 10^{-12}$ M

(log 4.2 = 0.62, log 4.8 = 6.8, log 6.2 = 0.80, log 7.5 = 0.88)

IE0070

Q.9 An ammonia-ammonium chloride buffer has a pH value of 9 with $[NH_3] = 0.25$. What will be the new pH if 500 ml 0.1 M KOH is added to 200 ml buffer solution ($K_b = 2 \times 10^{-5}$)

[log 2 = 0.3]

IE0071

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Q.10 A weak base (50.0mL) was titrated with 0.1 M HCl. The pH of the solution after the addition of 10.0 mL and 25.0 mL were found to be 9.84 and 9.24, respectively. Calculate K_b of the base and pH at the equivalence point. [log2 = 0.3]

IE0218

Q.11 A weak acid (50.0mL) was titrated with 0.1 M NaOH. The pH values when 10.0 mL and 25.0 mL of base have been added are found to be 4.16 and 4.76, respectively. Calculate K_a of the acid and pH at the equivalence point. [log2 = 0.3]

IE0219

Q.12 10 ml of 0.1M weak acid HA($k_a = 10^{-5}$) is mixed with 10 ml 0.2M HCl and 10 ml 0.1M NaOH. Find the value of $[A^-]$ in the resulting solution.

IE0072

Q.13 150 ml of 0.5 M HCN ($Ka = 3.75 \times 10^{-9}$) was reacted with 1.5 M KOH for complete neutralisation. What will be molarity of HCN at equilibrium.

IE0073

Q.14 The indicator phenol red is half in the ionic form when pH is 7.2. If the ratio of the undissociated form to the ionic form is 1 : 5, find the pH of the solution. With the same pH for solution, if indicator is altered such that the ratio of undissociated form to dissociated form becomes 1 : 4, find the pH when 50 % of the new indicator is in ionic form. [log2 = 0.3]

IE0220

Q.15 How much AgBr could dissolve in 1.0 L of 0.40 M NH_3 ? Assume that $Ag(NH_3)_2^+$ is the only complex formed. $[K_f(Ag(NH_3)_2^+) = 1 \times 10^8 ; K_{sp}(AgBr) = 5 \times 10^{-13}]$

$[\sqrt{50} \simeq 7]$

IE0074

Q.16 Calculate solubility of PbI_2 ($K_{sp} = 1.4 \times 10^{-8}$) in water at 25°, which is 90% dissociated.

$\left(\frac{1.4}{(0.81)(3.6)}\right)^{1/3} = 0.78$

IE0075

Q.17 A recent investigation of the complexation of SCN^- with Fe^{3+} led to 130, 16, and 1.0 for K_1, K_2, and K_3, respectively. What is the overall formation constant of $Fe(SCN)_3$ from its component ions, and what is the dissociation constant of $Fe(SCN)_3$ into its simplest ions on the basis of these data ?

IE0221

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EXERCISE # O-I

Single correct

Q.1 The conjugate acid of NH_2^- is

(A) NH_3 (B) NH_2OH (C) NH_4^+ (D) N_2H_4

IE0076

Q.2 Which of the following is not a Bronsted acid:-

(A) $CH_3NH_4^+$ (B) CH_3COO^- (C) H_2O (D) HSO_4^-

IE0077

Q.3 In the reaction

$HNO_3 + H_2O \rightleftharpoons H_3O^+ + NO_3^-$, the conjugate base of HNO_3 is :-

(A) H_2O (B) H_3O^+ (C) NO_3^- (D) H_3O^+ and NO_3^-

IE0078

Q.4 Out of the following, amphiprotic species in aqueous medium are

I: HPO_3^{2-} II OH^- III $H_2PO_4^-$ IV HCO_3^-

(A) I, III, IV (B) I and III (C) III and IV (D) All

IE0079

Q.5 When ammonia is added to water, it decreases the concentration of which of the following ion

(A) OH^- (B) H_3O^+ (C) NH_4^+ (D) NH_4^+ & OH^-

IE0080

Q.6 Which of the following pair is Lewis acid & Lewis base & Product of these is also Lewis base

(A) BF_3 , NH_3 (B) $SiCl_4$, $2Cl^-$ (C) $CH_3^{\oplus}$, $^{\ominus}OC_2H_5$ (D) All of these

IE0081

Q.7 Ionic product of water will increase, if :-

(A) Pressure is decreased (B) H^+ is added

(C) OH^- is increased (D) Temperature is increased

IE0082

Q.8 At 60°C, pure water has $[H_3O^+]=10^{-6.7}$mol/lit. what is the value of K_W at 60°C :-

(A) 10^{-6} (B) 10^{-12} (C) 10^{-67} (D) $10^{-13.4}$

IE0083

Q.9 Liquid NH_3 ionises to a slight extent. At a certain temperature its self ionization constant $K_{SIC(NH_3)} = 10^{-30}$. The number of NH_4^+ ions present per 100 cm^3 of pure liquid are

(A) 10^{-15} (B) 6.022×10^8 (C) 6.022×10^7 (D) None

IE0084

Q.10 The pH of solution is increased from 3 to 6. Its H^+ ion conc. will be :-

(A) Reduced to half (B) Doubled

(C) Reduced by 1000 times (D) Increased by 1000 times

IE0085

Q.11. pOH of [**1/200**] mol/m^3 H_2SO_4 (aq.) solution at 25ºC is-

(A) 2 (B) 5 (C) 9 (D) 12

IE0086

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Q.12 Degree of dissociation of 0.1 N CH_3COOH is :- (Dissociation constant = 1×10^{-5})
(A) 10^{-5} (B) 10^{-4} (C) 10^{-3} (D) 10^{-2}
IE0087

Q.13 The pH of a 0.02 M ammonia solution which is 5% ionised will be :-
(A) 2 (B) 11 (C) 5 (D) 7
IE0088

Q.14 The pH of an aqueous solution of 1.0 M solution of a weak monoprotic acid which is 1% ionised is
(A) 1 (B) 2 (C) 3 (D) 11
IE0089

Q.15 The concentration of $[H^+]$ and concentration of $[OH]^-$ of a 0.1 M aqueous solution of 2% ionised weak acid is [ionic product of water $=1 \times 10^{-14}$]
(A) 0.02×10^{-3} M and 5×10^{-11} M (B) 1×10^{-3} M and 3×10^{-11} M
(C) 2×10^{-3} M and 5×10^{-12} M (D) 3×10^{-2} M and 4×10^{-13} M
IE0090

Q.16 What is the quantity of NaOH present in 250 cc of the solution, so that it gives a pH = 13 :-
(A) 10^{-13} g (B) 10^{-1} g (C) 1.0 g (D) 4.0 g
IE0091

Q.17 An aqueous solution of HCl is 10^{-9} M HCl. The pH of the solution should be:-
(A) 9 (B) Between 6 and 7 (C) 7 (D) Unpredictable
IE0092

Q.18. The moles of H^+ from H_2O in a 1*l*, $\sqrt{5} \times 10^{-7}$ M HCl solution at 25°C, is
$(\sqrt{5} = 2.23)$
(A) 10^{-7} (B) 6.85×10^{-8} (C) 3.85×10^{-8} (D) 10^{-8}
IE0093

Q.19 Which one of the following has highest pH:-
(A) Distilled water (B) 1 M NH_3
(C) 1 M NaOH (D) Water saturated with chlorine
IE0094

Q.20 8 gm NaOH and 4.9 gm H_2SO_4 are present in one litre of the solution. What is its pH
(A) 1 (B) 13 (C) 12 (D) 2
IE0095

Q.21 10 ml of $\frac{M}{200}$ H_2SO_4 is mixed with 40 ml of $\frac{M}{200}$ H_2SO_4. The pH of the resulting solution is
(A) 1 (B) 2 (C) 2.3 (D) none of these
IE0096

Q.22 Which of the following solution will have pH close to 1.0?
(A) 100 ml of M/100 HCl + 100 ml of M/10 NaOH
(B) 55 ml of M/10 HCl + 45 ml of M/10 NaOH
(C) 10 ml of M/10 HCl + 90 ml of M/10 NaOH
(D) 75 ml of M/5 HCl + 25 ml of M/5 NaOH
IE0097

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Q.23 A solution with pH 2.0 is more acidic than the one with pH 6.0 by a factor of:
(A) 3 (B) 4 (C) 3000 (D) 10000
IE0098

Q.24 The first and second dissociation constants of an acid H_2A are 1.0×10^{-5} and 5.0×10^{-10} respectively. The overall dissociation constant of the acid will be :
(A) 5.0×10^{-5} (B) 5.0×10^{15} (C) 5.0×10^{-15} (D) 0.2×10^{5}
IE0099

Q.25 If pK_b for fluoride ion at 25° C is 10.4, the ionisation constant of hydrofluoric acid in water at this temperature is :
(A) 4×10^{-11} (B) 3×10^{-3} (C) 2.5×10^{-4} (D) 2×10^{-2}
IE0100

Q.26 pH of an aqueous solution of NaCl at 85°C should be
(A) 7 (B) > 7 (C) < 7 (D) 0
IE0101

Q.27 1 cc of 0.1 N HCl is added to 99 cc solution of NaCl. The pH of the resulting solution will be
(A) 7 (B) 3 (C) 4 (D) 1
IE0102

Q.28 The degree of hydrolysis of a salt of weak acid and weak base in it's 0.1 M solution is found to be 50%. If the molarity of the solution is 0.2 M, the percentage hydrolysis of the salt should be
(A) 100% (B) 50% (C) 25% (D) none of these
IE0103

Q.29 What is the percentage hydrolysis of NaCN in N/80 solution when the dissociation constant for HCN is 2×10^{-9} and $K_w = 1.0 \times 10^{-14}$
(A) 2 (B) 5.26 (C) 8.2 (D) 9.6
IE0104

Q.30 The compound whose 0.1 M solution is basic is
(A) Ammonium acetate (B) Ammonium chloride
(C) Ammonium sulphate (D) Sodium acetate
IE0105

Q.31 If equilibrium constant of
$CH_3COOH + H_2O \rightleftharpoons CH_3COO^- + H_3O^+$
is 1.8×10^{-5}, equilibrium constant for
$CH_3COOH + OH^- \rightleftharpoons CH_3COO^- + H_2O$ is
(A) 1.8×10^{-9} (B) 1.8×10^{9} (C) 5.55×10^{-9} (D) 5.55×10^{10}
IE0106

Q.32 The pK_a of a weak acid, HA, is 4.80. The pK_b of a weak base, BOH, is 4.78. The pH of an aqueous solution of the corresponding salt, BA, will be :
(A) 8.58 (B) 4.79 (C) 7.01 (D) 9.22
IE0107

Q.33 The highest pH value is of :-
(A) 0.1 M NaCl (B) 0.1 M NH_4Cl
(C) 0.1 M CH_3COONa (D) 0.1 M CH_3COONH_4
IE0108

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Q.34 pH of K_2S solution is:-
(A) 7 (B) Less than 7 (C) More than 7 (D) 0

IE0109

Q.35 Degree of Hydrolysis of $\frac{N}{100}$ solution of KCN is (Given Ka = 1.6×10^{-9})
(A) 2.5×10^{-3} (B) 2.5×10^{-2} (C) 2.5×10^{-4} (D) 2.5×10^{-5}

IE0110

Q.36 A solution of $FeCl_3$ in water acts as acidic due to :-
(A) Acidic impurities (B) Ionisation (C) Hydrolysis of Fe^{3+} (D) Dissociation

IE0111

Q.37 If 40 ml of 0.2 M KOH is added to 160 ml of 0.1 M HCOOH [$K_a = 2\times10^{-4}$], the pOH of the resulting solution is
(A) 3.4 (B) 3.7 (C) 7 (D) 10.3

IE0112

Q.38 1 M NaCl and 1M HCl are present in an aqueous solution. The solution is
(A) not a buffer solution and with pH < 7
(B) not a buffer solution with pH > 7
(C) a buffer solution with pH < 7
(D) a buffer solution with pH > 7

IE0113

Q.39 The pK_a of a weak acid (HA) is 4.5. The pOH of an aqueous buffered solution of HA in which 50% of the acid is ionized is :
(A) 4.5 (B) 2.5 (C) 9.5 (D) 7.0

IE0114

Q.40 To a 50 ml. of 0.05M formic acid, how much volume of 0.10M sodium formate must be added to get a buffer solution of pH = 4.0 ?
(pK_a of the acid is 3.7) (log2 = 0.3)
(A) 40 ml. (B) 4 ml. (C) 50 ml. (D) 100 ml.

IE0115

Q.41 Which can act as buffer :-
(A) NH_4OH + NaOH
(B) HCOOH + HCl
(C) 40 ml. of 0.1 M NaCN + 20 ml. of 0.1 M HCl
(D) All of them

IE0116

Q.42 If equal volume of 0.05 M ammonium hydroxide solution is dissolved in 0.001 M ammonium chloride solution. What will be the OH^- ion concentration of this solution :
$K_b(NH_4OH) = 1.8 \times 10^{-5}$
(A) 3.0×10^{-3} (B) 4.6×10^{-4} (C) 9.0×10^{-3} (D) 9.0×10^{-4}

IE0117

Q.43 Calculate the pH of a buffer prepared by mixing 600 cc of 0.6 M NH_3 and 400 cc of 0.5 M NH_4Cl. K_b for $NH_3 = 1.8 \times 10^{-5}$, (log 1.8 = 0.26)
(A) 11.3 (B) 9.0 (C) 9.52 (D) 5

IE0118

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Q.44 pK_b for NH_4OH at certain temperature is 4.74. The pH of basic buffer containing equimolar concentration of NH_4OH and NH_4Cl will be:-
(A) 7.74 (B) 4.74 (C) 2.37 (D) 9.26
IE0119

Q.45 On addition of NaOH to CH_3COOH solution, 60% of the acid is neutralised. If pK_a of CH_3COOH is 4.7 then the pH of the resulting solution is :-
(A) More than 4.7 but less than 5.0 (B) Less than 4.7 but more than 4.0
(C) More than 5.0 (D) Remains unchanged
IE0120

Q.46 Henderson equation $pH - pK_a = 5$ will be applicable to an acidic buffer when :-
(A) [Acid] = [Conjugate base] (B) [Acid] $\times 10^5$ = [Conjugate base]
(C) [Acid] = [Conjugate base] $\times 10^5$ (D) [acid] = 2 [conjugate base]
IE0121

Q.47 What amount of sodium propanoate should be added to one litre of an aqueous solution containing 0.02 mole of propanoic acid ($K_a = 3 \times 10^{-5}$ at 25°C) to obtain a buffer solution of pH 4.7
(A) 4.52×10^{-2} mol (B) 3.52×10^{-2} mol (C) 2.52×10^{-2} mol (D) 3×10^{-2} mol
IE0122

Q.48 In a buffer solution the ratio of concentration of NH_4Cl and NH_4OH is 1 : 1. When it changes in 2 : 1, what will be the value of pH of buffer ?
(A) Increase (B) Decrease (C) No effect (D) None
IE0123

Q.49 The buffer solution play an important role in :-
(A) Increasing the pH value (B) Decreasing the pH value
(C) Keeping the pH constant (D) Solution will be neutral
IE0124

Q.50 The total number of different kind of acidic buffers obtained during the titration of H_3PO_4 with NaOH are :
(A) 3 (B) 1 (C) 2 (D) 0
IE0125

Q.51 Which of the following solutions does not act as buffer :
(A) $H_3PO_4 + NaH_2PO_4$ (B) $NaHCO_3 + H_2CO_3$
(C) $NH_4Cl + HCl$ (D) $CH_3COOH + CH_3COONa$
IE0126

Q.52 Half of the formic acid solution is neutralised on addition of a KOH solution to it. If K_a (HCOOH) $= 2 \times 10^{-4}$ then pH of the solution is : - (log 2 = 0.3010)
(A) 3.6990 (B) 10.3010 (C) 3.85 (D) 4.3010
IE0127

Q.53 When 0.02 moles of NaOH are added to a litre of buffer solution, its pH changes from 5.75 to 5.80. What is its buffer capacity :-
(A) 0.4 (B) 0.05 (C) – 0.05 (D) 2.5
IE0128

Q.54 Calculate pH when 100 ml of 0.2M NaOH is reacted with 100 ml of 0.2 M CH_3COOH ($K_a = 10^{-5}$)
(A) 9 (B) 7 (C) 5 (D) 2
IE0129

Q.55 At 90°C , pure water has $[H^+] = 10^{-6}$ M, if 100 ml of 0.2 M HNO_3 is added to 20 ml of 1 M NaOH at 90°C then pH of the resulting solution will be
(A) 5 (B) 6 (C) 7 (D) None of these
IE0130

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Q.56 When 20 ml of $\frac{M}{20}$ NaOH are added to 10 ml of $\frac{M}{10}$ HCl, the resulting solution will:-

(A) Turn blue litmus red
(B) Turn phenolphthalein solution pink colour
(C) Turn methyl orange red
(D) Will have no effect on either red or blue litmus

IE0131

Q.57 The rapid change of pH near the stoichiometric point of an acid–base titration is the basis of indicator detection. pH of the solution is related to ratio of the concentrations of the conjugate acid (HIn) and base (In^-) forms of the indicator by the expression :-

(A) $\log \frac{[HIn]}{[In^-]} = pK_{In} - pH$ (B) $\log \frac{[HIn]}{[In^-]} = pH - pK_{In}$

(C) $\log \frac{[In^-]}{[HIn]} = pH + pK_{In}$ (D) $\log \frac{[In^-]}{[HIn]} = pK_{In} - pH$

IE0132

Q.58 Calculate the pH range in which an acid indicator with K_{acid} (indicator) $= 1.0 \times 10^{-5}$ changes colour when the concentration of the indicator is 1×10^{-3}M.

(A) 5 ± 1 (B) 11 ± 1 (C) 3 ± 1 (D) 8 ± 1

IE0133

Q.59 In what pH range will a 1×10^{-4} M solution of an indicator will K_b (indicator) $= 1 \times 10^{-11}$ change colour?

(A) 7.0 ± 1 (B) 3.0 ± 1 (C) 5.5 ± 1 (D) 11.0 ± 1

IE0134

Q.60 Indicator which is used in the titration of CH_3COOH & NaOH :–

(A) Methyl orange (B) Methyl red (C) Phenolphthalein (D) Litmus

IE0135

Q.61 Phenolphthalein is a :-

(A) Strong acid (B) Strong base (C) Weak base (D) Weak acid

IE0136

Q.62 pH–range of Methyl red indicator is :–

(A) 4·2–6·2 (B) 6·8–10·8 (C) 8 – 9·6 (D) 6·8 – 8·2

IE0137

Q.63 In the volumetric estimation of HCl, if we make use of phenolphthalein as an indicator, which base is unsuitable for the titration :-

(A) NaOH (B) RbOH (C) KOH (D) NH_4OH

IE0138

Q.64 Phenolphthalein does not act as an indicator for the titration between :-

(A) KOH and H_2SO_4 (B) NaOH and CH_3COOH
(C) Oxalic acid and $KMnO_4$ (D) $Ba(OH)_2$ and HCl

IE0139

Q.65 For weak acid and strong base titration, the indicator used is :-

(A) Potassium di-chromate (B) Methyl orange
(C) Litmus (D) Phenolphthalein

IE0140

Q.66 The solubility of A_2X_3 is y mol dm^{-3}. Its solubility product is

(A) $6\ y^2$ (B) $64\ y^4$ (C) $36\ y^5$ (D) $108\ y^5$

IE0141

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Q.67 If K_{sp} for $HgSO_4$ is 6.4×10^{-5}, then solubility of this substance in mole per m^3 is
(A) 8×10^{-3} (B) 6.4×10^{-5} (C) 8×10^{-6} (D) 8

IE0142

Q.68 If the solubility of AgCl (formula mass=143) in water at 25°C is 1.43×10^{-4} gm/100 ml of solution then the value of K_{sp} will be :–
(A) 1×10^{-5} (B) 2×10^{-5} (C) 1×10^{-10} (D) 2×10^{-10}

IE0143

Q.69 One litre of saturated solution of $CaCO_3$ is evaporated to dryness, 7.0 g of residue is left. The solubility product for $CaCO_3$ is:-
(A) 4.9×10^{-3} (B) 4.9×10^{-5} (C) 4.9×10^{-9} (D) 4.9×10^{-7}

IE0144

Q.70 A_3B_2 is a sparingly soluble salt of molar mass M (g mol^{-1}) and solubility x g lit^{-1}. The ratio of the molar concentration of B^{3-} to the solubility product of the salt is

(A) $108 \frac{x^5}{M^5}$ (B) $\frac{1}{108}\frac{M^4}{x^4}$ (C) $\frac{1}{54}\frac{M^4}{x^4}$ (D) None

IE0145

Q.71. Solubility of Ag_2CrO_4 ($K_{sp} = 4 \times 10^{-13}$) in 0.1 M K_2CrO_4 solution will be :–
(A) 10^{-3} M (B) 10^{-6} M (C) 4×10^{-6} M (D) 5×10^{-7} M

IE0146

Q.72. How many times solubility of CaF_2 is decreased in 4×10^{-3} M KF (aq.) solution as compare to pure water at 25°C. Given K_{sp} (CaF_2) = 3.2×10^{-11}
(A) 50 (B) 100 (C) 500 (D) 1000

IE0147

Q.73 At 30^0C, In which of the one litre solution, the solubility of Ag_2CO_3 (solubility product = 8×10^{-12}) will be maximum :-
(A) 0.05 M Na_2CO_3 (B) Pure water
(C) 0.05 M $AgNO_3$ (D) 0.05 M NH_3

IE0148

Q.74 What will happen if the pH of the solution of 0.001 M $Mg(NO_3)_2$ solution is adjusted to pH = 9 (K_{sp} of $Mg(OH)_2$ = 8.9×10^{-12})
(A) ppt will take place
(B) ppt will not take place
(C) Solution will be saturated
(D) None of these

IE0149

Q.75 Na_3PO_4 which should be added in 10 L of 1.0×10^{-5} M - $BaCl_2$ solution without any precipitation of $Ba_3(PO_4)_2$ is [Ksp of $Ba_3(PO_4)_2$] = 4×10^{-23}
(A) 2×10^{-4} gm (B) 0.328 gm (C) 0.164 gm (D) 0.82 gm

IE0150

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EXERCISE # O-II

Single correct :

Q.1 The pH of the solution produced when an aqueous solution of strong acid pH 5 is mixed with equal volume of an aqueous solution of strong acid of pH 3 is :-

(A) 3.3 (B) 3.5 (C) 4.5 (D) 4.0 **IE0151**

Q.2 How many moles of HCl must be removed from 1 litre of aqueous HCl solution to change its pH from 2 to 3 :-

(A) 1 (B) 0.02 (C) 0.009 (D) 0.01 **IE0152**

Q.3 Which of the following is most soluble in water?

(A) MnS ($K_{sp}= 8\times10^{-37}$) (B) ZnS ($K_{sp}= 7\times10^{-16}$)

(C) Bi_2S_3 ($K_{sp}= 1\times10^{-72}$) (D) $Ag_3(PO_4)$ ($K_{sp}= 1.8\times10^{-18}$) **IE0153**

Q.4 Solubility of AgBr will be minimum in :-

(A) Pure water (B) 0.1 M $CaBr_2$ (C) 0.1 M NaBr (D) 0.1 M $AgNO_3$ **IE0154**

Q.5 pH of solution at first 1/4th equivalence point of Na_2CO_3 when titrated with HCl will be (for H_2CO_3 $K_{a_1} = 10^{-7}$; $K_{a_2} = 10^{-11}$)

(A) 7 + log 3 (B) 7 – log3 (C) 11 + log3 (D) 11 – log3 **IE0155**

Q.6 An acid-base indicator has a $K_a = 1.0 \times 10^{-5}$. The acid form of the indicator is red and the basic form is blue. Calculate the pH change required to change the colour of the indicator from 80% red to 80% blue.

(A) 1.20 (B) 0.80 (C) 0.20 (D) 1.40 **IE0156**

Assertion /Reason :

Q.7 **Statement-1** pH of 10^{-7} M NaOH solution is exist between 7 to 7.3 at 25°C.

Statement-2 Due to common ion effect ionization of water is reduced.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

IE0157

Q.8 **Statement-1** In general phenolphthalein is used as an indicator for the titration of weak acid (HA) against strong base (NaOH)

Statement-2 At equivalent point solution is basic.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

IE0222

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Q.9 **Statement-1 :** Moles of Sr^{2+} furnished by sparingly soluble substance $Sr(OH)_2$ decreases due to dilution.

Statement-2 : Solubility product constant of $Sr(OH)_2$ is not affected by dilution.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

IE0223

Q.10 **Statement-1** : On dilution of a concentrated solution of CH_3COOH , the concentration of $[H^+]$ decreases.

Statement-2 : Increase in volume is more than the increase in degree of ionisation.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

IE0158

Multiple correct :

Q.11 Which of the following is correct for 0.1 M BOH solution ($K_b = 10^{-5}$)

(A) pH of solution is 11

(B) OH^- concentration is 10^{-3} mol/L

(C) it's salt with HCl (i.e. BCl) form the acidic solution in water

(D) Phenolphthalein indicator can be used during the titration of BOH with HCl

IE0159

Q.12 For weak monobasic acid, HA, the dissociation constant is 2×10^{-6} , at 25ºC. Which of the following is/are correct regarding this acid? [log2= 0.3]

(A) $A^- + H_2O \rightleftharpoons HA + OH^-$; $K_{eq} = 5 \times 10^{-9}$

(B) The equilibrium constant for the reaction of HA with aq. NaOH is 2×10^8

(C) The pH of 0.1 M, HA solution is 3.35

(D) solution of A^- is basic

IE0160

Q.13 Select correct statement for 50ml 0.1M H_2A(aq.) solution ; $K_{a_1} = 10^{-5}$; $K_{a_2} = 10^{-8}$

(A) $[H^+] = 2[A^{2-}]$

(B) pH of 0.1M H_2A solution is 3

(C) In above H_2A solution when 5 milimoles of NaHA are added then pH increases by 2 units

(D) 50 ml of 0.1M NaOH required to neutralised completely 50 ml of 0.1M H_2A solution

IE0224

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Q.14 A solution containing 0.01M each of Pb^{2+}, Ag^{+}, Zn^{2+} & Cr^{3+} ion. If solid Na_2S is added slowly to the solution then correct statement is based on given data -

Precipitate	**PbS**	**Ag_2S**	**ZnS**	**Cr_2S_3**
Ksp	**10^{-11}**	**10^{-12}**	**10^{-8}**	**10^{-10}**

(A) Pb^{2+} will start precipitating first

(B) Cr^{3+} will start precipitating last

(C) Zn^{2+} will start precipitating before Ag^{+}

(D) When Zn^{2+} just starts precipitating then Pb^{2+} ion gets 99.9% precipitated

IE0225

Q.15 If K_1 & K_2 be first and second ionisation constant of H_3PO_4 and $K_1 >> K_2$ which is/are incorrect.

(A) $[H^+] = [H_2PO_4^-]$ (B) $[H^+] = \sqrt{K_1[H_3PO_4]}$

(C) $K_2 = [HPO_4^{--}]$ (D) $[H^+] = 3[PO_4^{3-}]$

IE0161

Q.16 10 ml. of a solution contains 0.1 M NH_4Cl + 0.01M NH_4OH. Which addition would not change the pH of solution :–

(A) Adding 1 ml. water

(B) Adding 5 ml. of 0.1 M NH_4Cl

(C) Adding 5 ml. of 0.1 M NH_4OH

(D) Adding 10 ml. of 0.1 M NH_4Cl

IE0162

Q.17 When equal volumes of the following solutions are mixed, precipitation of AgCl ($K_{sp} = 1.8 \times 10^{-10}$) will occur only with:

(A) 10^{-4} M (Ag^+) and 10^{-4} M (Cl^-)

(B) 10^{-5} M (Ag^+) and 10^{-5} M (Cl^-)

(C) 10^{-6} M (Ag^+) and 10^{-6} M (Cl^-)

(D) 10^{-10} M (Ag^+) and 10^{-10} M (Cl^-)

IE0163

Paragraph for Q.18 to Q.20

8 gm weak acid HX (molecular mass = 80) is dissolved is 100 ml water. ($K_a = 10^{-4}$)

Q.18 Find pH of solution-

(A) 3.3 (B) 2 (C) 2.3 (D) 3

IE0164

Q.19 If it is titrated with 0.25 M NaOH find pH at equivalence point (log5 = 0.7)

(A) 9.15 (B) 8.65 (C) 4.65 (D) 4.85

IE0165

Q.20 Find $[H^+]$ if 10^{-3} mol HCl is added to 100 ml original solution $\left(\sqrt{41} = 6.4 \;;\; \sqrt{5} = 2.24\right)$

(A) 0.62×10^{-2} (B) 1.62×10^{-2} (C) 2.7×10^{-2} (D) 0.27×10^{-2}

IE0166

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Paragraph for Question 21 & 22

A solution contains one mole each of HA & HB (both are weak acids) in one litre solution. Now one mole of NaOH is added to this solution so that both the acids are partially neutralised. Heat of neutralisation of HA & HB are –11.8 and –12.4 kcal per mole respectively and heat produced during partial neutralisation of HA & HB is –12.25 kcal.

Q.21 Mole ratio of neutralisation of HA & HB is -

(A) 1 : 4 (B) 1 : 2 (C) 1 : 3 (D) 1 : 5

IE0226

Q.22 pH of solution containing mixture of 1 mol of HA & 1 mole of HB in 1 litre solution is

$K_a(HA) = 1 \times 10^{-5} \cdot K_a(HB) = 9 \times 10^{-5}$

(A) 2 (B) 3 (C) 4 (D) 5

IE0227

MATCH THE COLUMN :

Q.23 Match the effect of addition of 1 M NaOH to 100 mL 1 M CH_3COOH (in Column I) with pH (in Column II) :

Column-I	**Column-II**
(A) 25 mL of NaOH	(P) pK_a
(B) 50 mL of NaOH	(Q) $pK_a + \log 3$
(C) 75 mL of NaOH	(R) $pK_a - \log 3$
(D) 100 mL of NaOH	(S) $\frac{1}{2}[pK_w + pK_a - \log 2]$

IE0167

Q.24

	Column-I **PH**		**Column-II** **Solution**
(A)	3	(P)	When equal volumes of 0.2M NH_4OH ($K_b = 10^{-5}$) & 0.2M HCl are mixed
(B)	5	(Q)	When equal volumes of 0.2M CH_3COONa & 0.2M HCl are mixed ($K_{a(CH_3COOH)} = 10^{-5}$)
(C)	8	(R)	0.1M Na_2HPO_4 (for H_3PO_4 ; $K_{a_1} = 10^{-4}$; $K_{a_2} = 10^{-6}$; $K_{a_3} = 10^{-10}$)
(D)	9	(S)	At 1st half equivalence point of H_2CO_3 when titrated against 0.1M NaOH , $K_{a_1} = 10^{-5}$, $K_{a_2} = 10^{-9}$
		(T)	$Mg(OH)_2$; $K_{sp} = 5 \times 10^{-16}$

IE0228

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\4-TD\01-TC-TD-02 & 1 Eq.-(PQ)\Eng\4-Ionic Equilibrium\02_Ex.p65

TABLE TYPE QUESTION :

Column-I (solution)	**Column-II (pH of solution)**	**Column-III (Introduction about solution)**
(A) CH_3COOH(0.2M, 1L) + NaOH (0.1M, 1 L)	(P) 1.3	(1) pH is determined by strong acid
(B) CH_3COOH(0.1M ,1 L) + HCl (0.1 M, 1 L)	(Q) 7	(2) Buffer solution at its maximum buffer capacity
(C) CH_3COOH(0.1M, 1 L) + NH_4OH (0.1M, 1 L)	(R) 9	(3) pH is determined by salt hydrolysis.
(D) NH_4Cl (200 ml, 0.1M) + NaOH (100 ml, 0.1M)	(S) 5	(4) pH is determined by buffer solution

(Given : $(K_a)_{CH_3COOH} = 10^{-5}$, $(K_b)_{NH_4OH} = 10^{-5}$)

Q.25 Which of the following is incorrectly matched

(A) A - S - 4 (B) B - P - 1 (C) D - R - 2 (D) C - Q - 1

IE0168

Q.26 If 0.15 mole NaOH is added in solution (B) of column-I then which of the following is correct

(A) S - 3 (B) S - 4 (C) R - 1 (D) P - 3

IE0169

Q.27 If 0.1mole HCl is added in solution (A) of column-I then pH of the resulting solution will be

(A) 7 (B) 13 (C) 3.0 (D) 1

IE0170

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\4-TD-01-TC-TD-02 & 1 Eq.-(PC)\Eng\4-Ionic Equilibrium\02_Ex.p65

EXERCISE # J-MAIN

Q.1 pK_a of a weak acid (HA) and pK_b of a weak base (BOH) are 3.2 and 3.4, respectively. The pH of their salt (AB) solution is **[JEE-MAIN(Offine)–2017]**

(1) 7.2 (2) 6.9 (3) 7.0 (4) 1.0

IE0182

Q.2 Addition of sodium hydroxide solution to a weak acid (HA) results in a buffer of pH 6. If ionisation constant of HA is 10^{-5}, the ratio of salt to acid concentration in the buffer solution will be : **[JEE-MAIN(Online)–2017]**

(1) 4 : 5 (2) 1 : 10 (3) 10 : 1 (4) 5 : 4

IE0183

Q.3 50 mL of 0.2 M ammonia solution is treated with 25 mL of 0.2 M HCl. If pK_b of ammonia solution is 4.75, the pH of the mixture will be:- **[JEE-MAIN(Online)–2017]**

(1) 8.25 (2) 4.75 (3) 9.25 (4) 3.75

IE0184

Q.4 Which of the following salts is the most basic in aqueous solution ? **[JEE-MAIN(Offine)–2018]**

(1) CH_3COOK (2) $FeCl_3$ (3) $Pb(CH_3COO)_2$ (4) $Al(CN)_3$

IE0185

Q.5 An alkali is titrated against an acid with methyl orange as indicator, which of the following is a correct combination ? **[JEE-MAIN(Offine)–2018]**

	Base	Acid	End point
(1)	Strong	Strong	Pinkish red to yellow
(2)	Weak	Strong	Yellow to pinkish red
(3)	Strong	Strong	Pink to colourless
(4)	Weak	Strong	Colourless to pink

IE0186

Q.6 An aqueous solution contains 0.10 M H_2S and 0.20 M HCl. If the equilibrium constants for the formation of HS^- from H_2S is 1.0×10^{-7} and that of S^{2-} from HS^- ions is 1.2×10^{-13} then the concentration of S^{2-} ions in aqueous solution is : **[JEE-MAIN(Offine)–2018]**

(1) 3×10^{-20} (2) 6×10^{-21} (3) 5×10^{-19} (4) 5×10^{-8}

IE0187

Q.7 A aqueous solution contains an unknown concentration of Ba^{2+}. When 50 mL of a 1 M solution of Na_2SO_4 is added, $BaSO_4$ just begins to precipitate. The final volume is 500 mL. The solubility product of $BaSO_4$ is 1×10^{-10}. What is the original concentration of Ba^{2+} ? **[JEE-MAIN(Offine)–2018]**

(1) 2×10^{-9} M (2) 1.1×10^{-9} M (3) 1.0×10^{-10} M (4) 5×10^{-9} M

IE0188

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\4-TD\01-TC-TD-02 & 1 Eq.-(PQ)\Eng\4-Ionic Equilibrium\02_Ex.p65

Q.8 Following four solutions are prepared by mixing different volumes of NaOH and HCl of different concentrations, pH of which one of them will be equal to 1 ? **[JEE-MAIN(Online)–2018]**

(1) $75mL\frac{M}{5}HCl + 25mL\frac{M}{5}NaOH$

(2) $100mL\frac{M}{10}HCl + 100mL\frac{M}{10}NaOH$

(3) $55mL\frac{M}{10}HCl + 45mL\frac{M}{10}NaOH$

(4) $60mL\frac{M}{10}HCl + 40mL\frac{M}{10}NaOH$

IE0189

Q.9 The minimum volume of water required to dissolve 0.1 g lead (II) chloride to get a saturated solution (K_{sp} of $PbCl_2 = 3.2 \times 10^{-8}$; atomic mass of Pb = 207 u) is : **[JEE-MAIN(Online)–2018]**

(1) 0.36 L (2) 0.18 L (3) 17.98 L (4)1.798 L

IE0190

Q.10 If K_{sp} of Ag_2CO_3 is 8×10^{-12}, the molar solubility of Ag_2CO_3 in 0.1M $AgNO_3$ is :

(1) 8×10^{-12} M (2) 8×10^{-10} M **[JEE-MAIN(Online)–2019**

(3) 8×10^{-11} M (4) 8×10^{-13} M

IE0240

Q.11 25 ml of the given HCl solution requires 30 mL of 0.1 M sodium carbonate solution. What is the volume of this HCl solution required to titrate 30 mL of 0.2 M aqueous NaOH solution?

[JEE-MAIN(Online)–2019]

(1) 25 mL (2) 50 mL (3) 12.5 mL (4) 75 mL

IE0241

Q.12 A mixture of 100 m mol of $Ca(OH)_2$ and 2g of sodium sulphate was dissolved in water and the volume was made up to 100 mL. The mass of calcium sulphate formed and the concentration of OH^- in resulting solution, respectively, are :

(Molar mass of $Ca(OH)_2$, Na_2SO_4 and $CaSO_4$ are 74, 143 and 136 g mol^{-1}, respectively; K_{sp} of $Ca(OH)_2$ is 5.5×10^{-6})

[JEE-MAIN(Online)–2019]

(1) 1.9 g, 0.14 mol L^{-1} (2) 13.6 g, 0.14 mol L^{-1}

(3) 1.9 g, 0.28 mol L^{-1} (4) 13.6 g, 0.28 mol L^{-1}

IE0242

Q.13 20 mL of 0.1 M H_2SO_4 solution is added to 30 mL of 0.2 M NH_4OH solution. The pH of the resultant mixture is : [pk_b of NH_4OH = 4.7]. **[JEE-MAIN(Online)–2019]**

(1) 9.4 (2) 5.0 (3) 9.0 (4) 5.2

IE0244

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\4-TD-01-TC-TD-02 & 1 Eq - (PQ)\Eng\4-Ionic Equilibrium\02_Ex.p65

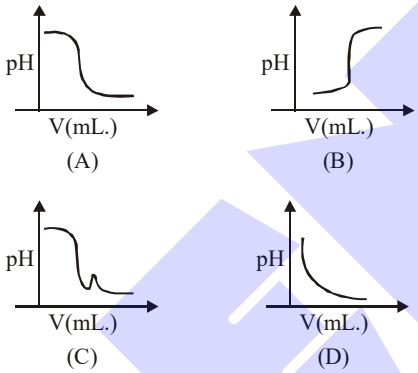
Q.14 If solubility product of $Zr_3(PO_4)_4$ is denoted by K_{sp} and its molar solubility is denoted by S, then which of the following relation between S and K_{sp} is correct **[JEE-MAIN(Online)–2019]**

(1) $S=\left(\frac{K_{sp}}{929}\right)^{1/9}$ (2) $S=\left(\frac{K_{sp}}{216}\right)^{1/7}$ (3) $S=\left(\frac{K_{sp}}{144}\right)^{1/6}$ (4) $S=\left(\frac{K_{sp}}{6912}\right)^{1/7}$

IE0245

Q.15 In an acid-base titration, 0.1 M HCl solution was added to the NaOH solution of unknown strength. Which of the following correctly shows the change of pH of the titraction mixture in this experiment?

[JEE-MAIN(Online)–2019]



(1) (A) (2) (C) (3) (D) (4) (B)

IE0246

Q.16 The pH of a 0.02M NH_4Cl solution will be **[JEE-MAIN(Online)–2019]**
[given $K_b(NH_4OH)=10^{-5}$ and log2=0.301]

(1) 4.65 (2) 5.35 (3) 4.35 (4) 2.65

IE0247

Q.17 The molar solubility of $Cd(OH)_2$ is 1.84×10^{-5} M in water. The expected solubility of $Cd(OH)_2$ in a buffer solution of pH = 12 is : **[JEE-MAIN(Online)–2019]**

(1) 6.23×10^{-11} M (2) 1.84×10^{-9} M (3) $\frac{2.49}{1.84}\times10^{-9}$ M (4) 2.49×10^{-10} M

IE0248

Q.18 Two solutions A and B, each of 100 L was made by dissolving 4g of NaOH and 9.8 g of H_2SO_4 in water, respectively. The pH of the resultant solution obtained from mixing 40 L of solution A and 10 L of solution B is_____. **[JEE-MAIN(Online)–2020]**

IE0249

Q.19 3g of acetic acid is added to 250 mL of 0.1 M HCl and the solution made up to 500 mL. To 20 mL of this solution $\frac{1}{2}$ mL of 5 M NaOH is added. The pH of the solution is **[JEE-MAIN(Online)–2020]**

[Given : pK_a of acetic acid = 4.75, molar mass of acetic acid = 60 g/mol, log3 = 0.4771]
Neglect any changes in volume

IE0250

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\v4-TD\01-TC-TD-02 & 1 Eq.-(PQ)\Eng\4-Ionic Equilibrium\02_Ex.p65

Q.20 For the following Assertion and Reason, the correct option is : **[JEE-MAIN(Online)–2020]**

Assertion : The pH of water increases with increase in temperature.

Reason : The dissociation of water into H^+ and OH^- is an exothermic reaction.

(1) Both assertion and reason are true, but the reason is not the correct explanation for the assertion.

(2) Both assertion and reason are false.

(3) Assertion is not true, but reason is true.

(4) Both assertion and reason are true, and the reason is the correct explanation for the assertion.

IE0252

Q.21 The K_{sp} for the following dissociation is 1.6×10^{-5} **[JEE-MAIN(Online)–2020]**

$$PbCl_{2(s)} \rightleftharpoons Pb^{2+}_{(aq)} + 2Cl^-_{(aq)}$$

Which of the following choices is correct for a mixture of 300 mL 0.134 M $Pb(NO_3)_2$ and 100 mL 0.4 M NaCl ?

(1) $Q < K_{sp}$ (2) $Q > K_{sp}$ (3) $Q = K_{sp}$ (4) Not enough data provided

IE0253

Q.22 A solution is 0.1 M in Cl^- and 0.001 M in CrO_4^{2-}. Solid $AgNO_3$ is gradually added to it

Assuming that the addition does not change in volume and $K_{sp}(AgCl) = 1.7 \times 10^{-10}$ M^2 and $K_{sp}(Ag_2CrO_4) = 1.9 \times 10^{-12}$ M^3.

Select correct statement from the following : **[JEE-MAIN(Online)–2021]**

(1) AgCl precipitates first because its K_{sp} is high.

(2) Ag_2CrO_4 precipitates first as its K_{sp} is low.

(3) Ag_2CrO_4 precipitates first because the amount of Ag^+ needed is low.

(4) AgCl will precipitate first as the amount of Ag^+ needed to precipitate is low.

IE0273

Q.23 The solubility of $Ca(OH)_2$ in water is : **[JEE-MAIN(Online)–2021]**

[Given : The solubility product of $Ca(OH)_2$ in water = 5.5×10^{-6}]

(1) 1.77×10^{-6} (2) 1.11×10^{-6} (3) 1.11×10^{-2} (4) 1.77×10^{-2}

IE0274

Q.24 The solubility of AgCN in a buffer solution of pH = 3 is x. The value of x is:

[Assume : No cyano complex is formed; $K_{sp}(AgCN) = 2.2 \times 10^{-16}$ and $K_a(HCN) = 6.2 \times 10^{-10}$]

(1) 0.625×10^{-6} (2) 1.9×10^{-5} (3) 2.2×10^{-16} (4) 1.6×10^{-6}

[JEE-MAIN(Online)–2021]

IE0275

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem Module\4-TD-01-TC-TD-02 & 1 Eq - (PQ)\Eng\4-Ionic Equilibrium\02_Ex.p65

Q.25 Assuming that $Ba(OH)_2$ is completely ionised in aqueous solution under the given conditions the concentration of H_3O^+ ions in 0.005 M aqueous solution of $Ba(OH)_2$ at 298 K is _____$\times 10^{-12}$ mol L^{-1}. (Nearest integer)

[JEE-MAIN(Online)–2021]

IE0276

Q.26 Sulphurous acid (H_2SO_3) has $Ka_1 = 1.7 \times 10^{-2}$ and $Ka_2 = 6.4 \times 10^{-8}$. The pH of 0.588 M H_2SO_3 is _____. (Round off to the Nearest Integer)

IE0277

Q.27 Two salts A_2X and MX have the same value of solubility product of 4.0×10^{-12}. The ratio of their molar solubilities i.e. $\frac{S(A_2X)}{S(MX)}$ = _____. (Round off to the Nearest Integer).

[JEE-MAIN(Online)–2021]

IE0278

Q.28 In order to prepare a buffer solution of pH 5.74, sodium acetate is added to acetic acid. If the concentration of acetic acid in the buffer is 1.0 M, the concentration of sodium acetate in the buffer is _____ M. (Round off to the Nearest Integer). **[JEE-MAIN(Online)–2021]**

[Given : pKa (acetic acid) = 4.74]

IE0279

Q.29 The solubility product of PbI_2 is 8.0×10^{-9}. The solubility of lead iodide in 0.1 molar solution of lead nitrate is $x \times 10^{-6}$ mol/L. The value of x is _____. (Rounded off to the nearest integer)

$[\text{Given} : \sqrt{2} = 1.41]$ **[JEE-MAIN(Online)–2021]**

IE0280

Q.30 The pH of ammonium phosphate solution, if pK_a of phosphoric acid and pk_b of ammonium hydroxide are 5.23 and 4.75 respectively, is _____.(Nearest integer) **[JEE-MAIN(Online)–2021]**

IE0281

Q.31 The pH of a solution obtained by mixing 50 mL of 1 M HCl and 30 mL of 1 M NaOH is $x \times 10^{-4}$. The value of x is _____. (Nearest integer) **[JEE-MAIN(Online)–2021]**

[log 2.5 = 0.3979]

IE0282

Q.32 A_3B_2 is a sparingly soluble salt of molar mass M (g mol^{-1}) and solubility x g L^{-1} . The solubility product satisfies Ksp = $a\left(\frac{x}{M}\right)^5$. The value of a is _____. (Integer answer)

[JEE-MAIN(Online)–2021]

IE0283

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\4-TD\01-TC-TD-02 & 1 Eq.-(PQ)\Eng\4.Ionic Equilibrium\02_Ex.p65

EXERCISE # J-ADVANCED

Q.1 A solution which is 10^{-3} M each in Mn^{2+}, Fe^{2+}, Zn^{2+} and Hg^{2+} is treated with 10^{-16}M sulphide ion. If K_{sp}, MnS, FeS, ZnS and HgS are 10^{-15}, 10^{-23}, 10^{-20} and 10^{-54} respectively, which one will precipitate first ? **[JEE 2003]**

(A) FeS (B) MnS (C) HgS (D) ZnS

IE0262

Q.2 0.1 M of HA is titrated with 0.1 M NaOH, calculate the pH at end point. Given Ka(HA) = 5×10^{-6} and $\alpha << 1$. **[JEE 2004]**

IE0264

Q.3 HX is a weak acid ($K_a = 10^{-5}$). It forms a salt NaX (0.1 M) on reacting with caustic soda. The degree of hydrolysis of NaX is

(A) 0.01% (B) 0.0001% (C) 0.1% (D) 0.5% **[JEE 2004]**

IE0265

Q.4 CH_3NH_2 (0.1 mole, $K_b = 5 \times 10^{-4}$) is added to 0.08 moles of HCl and the solution is diluted to one litre, resulting hydrogen ion concentration is

(A) 1.6×10^{-11} (B) 8×10^{-11} (C) 5×10^{-5} (D) 2×10^{-2} **[JEE 2005]**

IE0266

Q.5 If $Ag^+ + NH_3 \rightleftharpoons [Ag(NH_3)]^+$; $K_1 = 1.6 \times 10^3$ and **[JEE 2006]**

$[Ag(NH_3)]^+ + NH_3 \rightleftharpoons [Ag(NH_3)_2]^+$; $K_2 = 6.8 \times 10^3$.

The formation constant of $[Ag(NH_3)_2]^+$ is :

(A) 6.08×10^{-6} (B) 6.8×10^{-6} (C) 1.6×10^3 (D) 1.088×10^7

IE0267

Q.6 The species present in solution when CO_2 is dissolved in water :

(A) CO_2, H_2CO_3, HCO_3^-, CO_3^{2-} (B) H_2CO_3, CO_3^{2-} **[JEE 2006]**

(C) CO_3^{2-}, HCO_3^- (D) CO_2, H_2CO_3

IE0268

Q.7 2.5 mL of $\frac{2}{5}$ M weak monoacidic base ($K_b = 1 \times 10^{-12}$ at 25°C) is titrated with $\frac{2}{15}$ M HCl in water at 25°C. The concentration of H^+ at equivalence point is **[JEE 2008]**

($K_w = 1 \times 10^{-14}$ at 25°C)

(A) 3.7×10^{-13} M (B) 3.2×10^{-7} M (C) 3.2×10^{-2} M (D) 2.7×10^{-2} M

IE0269

Q.8 Solubility product constants (K_{SP}) of salts of types MX, MX_2 and M_3X at temperature 'T' are 4.0×10^{-8}, 3.2×10^{-14} and 2.7×10^{-15}, respectively. Solubilities (mol dm^{-3}) of the salts at temperature 'T' are in the order :

(A) $MX > MX_2 > M_3X$ (B) $M_3X > MX_2 > MX$ **[JEE 2008]**

(C) $MX_2 > M_3X > MX$ (D) $MX > M_3X > MX_2$

IE0270

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\4-TD-01-TC-TD-02 & Eq-(PQ)\Eng\4-Ionic Equilibrium\02_Ex.p65

Q.9 The dissociation constant of a substituted benzoic acid at 25°C is 1.0×10^{-4}. The pH of a 0.01 M solution of its sodium salt is **[JEE 2009]**

IE0271

Q.10 Aqueous solutions of HNO_3, KOH, CH_3COOH and CH_3COONa of identical concentrations are provided. The pair(s) of solutions which form a buffer upon mixing is(are) –

(A) HNO_3 and CH_3COOH (B) KOH and CH_3COONa

(C) HNO_3 and CH_3COONa (D) CH_3COOH and CH_3COONa

IE0272

Q.11 In 1 L saturated solution of AgCl [K_{sp}(AgCl) = 1.6×10^{-10}], 0.1 mol of CuCl [K_{sp}(CuCl) = 1.0×10^{-6}] is added. The resultant concentration of Ag^+ in the solution is 1.6×10^{-x}. The value of 'x' is. **[JEE -2011]**

IE0196

Q.12 The initial rate of hydrolysis of methyl acetate (1M) by a weak acid (HA, 1M) is $1/100^{th}$ of that of a strong acid (HX, 1M), at 25°C. The K_a of HA is **[JEE 2013]**

(A) 1×10^{-4} (B) 1×10^{-5} (C) 1×10^{-6} (D) 1×10^{-3}

IE0197

Q.13 The K_{sp} of Ag_2CrO_4 is 1.1×10^{-12} at 298 K. The solubility (in mol/L) of Ag_2CrO_4 in a 0.1 M $AgNO_3$ solution is **[JEE 2013]**

(A) 1.1×10^{-11} (B) 1.1×10^{-10} (C) 1.1×10^{-12} (D) 1.1×10^{-9}

IE0198

Paragraph For Questions 14 and 15

When 100 mL of 1.0 M HCl was mixed with 100 mL of 1.0 M NaOH in an insulated beaker at constant pressure, a temperature increase of 5.7 °C was measured for the beaker and its contents. **(Expt-1).** Because the enthalpy of neutralisation of a strong acid with a strong base is a constant (–57.0 $kJmol^{-1}$), this experiment could be used to measure the calorimeter constant. In a second experiment **(Expt-2)**, 100 mL of 2.0 M acetic acid ($K_a = 2.0 \times 10^{-5}$) was mixed with 100 mL of 1.0M NaOH (under identical conditions to **(Expt-1))** where a temperature rise of 5.6 °C was measured. (Consider heat capacity of all solutions as 4.2 $Jg^{-1}K^{-1}$ and density of all solutions as 1.0 g mL^{-1})

Q.14 Enthalpy of dissociation (in kJ mol^{-1}) of acetic acid obtained from the **Expt-2** is **[JEE 2015]**

(A) 1.0 (B) 10.0 (C) 24.5 (D) 51.4

IE0199

Q.15 The pH of the solution after **Expt-2**

(A) 2.8 (B) 4.7 (C) 5.0 (D) 7.0

IE0200

Q.16 The solubility of a salt of weak acid(AB) at pH = 3 is $Y \times 10^{-3}$ mol L^{-1}. The value of Y is___.

(Given that the value of solubility product of AB (K_{sp}) = 2×10^{-10} and the value of ionization constant of HB(K_a) = 1×10^{-8}) **[JEE 2018]**

IE0201

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\4-TD\01-TC-TD-02 & 1 Eq.-(PQ)\Eng\4.Ionic Equilibrium\02_Ex.p65

Q.17 For the following reaction, the equilibrium constant K_c at 298 K is 1.6×10^{17}.

$Fe^{2+}(aq) + S^{2-}(aq) \rightleftharpoons FeS(s)$ **[JEE 2019]**

When equal volumes of 0.06 M $Fe^{2+}(aq)$ and 0.2 M $S^{2-}(aq)$ solutions are mixed, the equilibrium concentration of $Fe^{2+}(aq)$ is found to be $\mathbf{Y} \times 10^{-17}$ **M**. The value of Y is ————

IE0284

Q.18 5.00 mL of 0.10 M oxalic acid solution taken in a conical flask is titrated against NaOH from a burette using phenolphthalein indicator. The volume of NaOH required for the appearance of permanent faint pink color is tabulated below for five experiments. What is the concentration, in molarity, of the NaOH solution ? **[JEE 2020]**

Exp. No.	Vol. of NaOH (mL)
1	12.5
2	10.5
3	9.0
4	9.0
5	9.0

IE0285

Q.19 An acidified solution of 0.05 M Zn^{2+} is saturated with 0.1 M H_2S. What is the minimum molar concentration (M) of H^+ required to prevent the precipitation of ZnS ?
Use K_{sp} (ZnS) = 1.25×10^{-22} and overall dissociation constant of H_2S , **[JEE 2020]**
$K_{NET} = K_1K_2 = 1 \times 10^{-21}$

IE0286

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\4-TD-01-TC-TD-02 & IEq-(PC)\Eng\4-Ionic Equilibrium\02_Ex.p65

ANSWER KEY

EXERCISE # S-I

Q.1 6.022×10^7 Q.2 (i) 6.53 ; (ii) (a) Basic , (b) Acidic

Q.3 6.8

Q.4 (a) 1, (b) 2.87, (c) 11.13 (d) 6.97, (e) 7, (f) 6 , (g) 6.97, (h) 11.30 (i) 9 , (j) 3

Q.5 (a) $K_a = 10^{-8}$, (b) $K_b = 10^{-6}$ Q.6 10

Q.7 173.2 : 1 Q.8 0.009

Q.9 2.32×10^{-8} M Q.10 1.1×10^{-3} M

Q.11 1.11×10^{-4} Q.12 4.7

Q.13 3.3 Q.14 (a) 0.522 , (b) 2.522

Q.15. (1)

Q.16 $[H^+] = 1.65 \times 10^{-2}$M, $[CHCl_2COO^-] = 6.5 \times 10^{-3}$M

Q.17 $[H^+] = 3 \times 10^{-3}$ M, $[CH_3COO^-] = 1.2 \times 10^{-3}$M, $[C_7H_5O_2^-] = 1.8 \times 10^{-3}$M

Q.18 0.027 M, 0.073 M, 0.027 M, 10^{-5} M

Q.19 $[H^+] = [H_2PO_4^-] = 2.7 \times 10^{-3}$M, $[HPO_4^{2-}] = 10^{-8}$M, $[PO_4^{3-}] = 3.7 \times 10^{-19}$M

Q.20 11.3 Q.21 10^{-5}M

Q.22 pH = 4.5 Q.23 $K_b = 6.25 \times 10^{-10}$

Q.24 0.56%, pH = 7 Q.25 1.667%

Q.26 0.25 % Q.27 10^{-6} ; 10^{-8}

Q.28 8.3 Q.29 4.19

Q.30 (a) 6, (b) 1×10^{-5} Q.31 9.0

Q.32 9.56 Q.33 5.04

Q.34 0.05 mol Q.35 $[OH^-] = 9.0 \times 10^{-6}$M

Q.36 (10.1) Q.37 4.74

Q.38 9.56 Q.39 8.7782

Q.40 (3.33) Q.41 10^{-5} M

Q.42 8.7 , $[H_3O^+] = 2 \times 10^{-9}$M Q.43 5 , 10^{-5} M

Q.44 (i) 2.85, (ii) 4.0969, (iii) 4.5229, (iv) 4.699, (v) 5.301, (vi) 8.699

Q.45 $[HI_n] = 80$ % Q.46 85.71%

Q.47 (b), (c) Q.48 $\Delta pH = 0.954$

Q.49 QX_2 is more soluble Q.50 4×10^{-8}

Q.51 5×10^{-11} Q.52 3.2×10^{-11}

Q.53 2.56×10^{-16} Q.54 1.0×10^{-5} mol/lit

Q.55 5×10^{-10} M Q.56 5×10^{-7}

Q.57 4×10^{-7} mol/L AgBr, 1.6×10^{-6} mol/L AgSCN

Q.58 $[F^-] = 3 \times 10^{-3}$M Q.59 8×10^{-3} M

Q.60 (i) 2×10^{-9} , (ii) 8×10^{-3} Q.61 $S = 2 \times 10^{-4}$ M, pH = 8.0

Q.62 4×10^{-2}M Q.63 (a) no precipitation will occur, (b) a precipitate will form

Q.64 No. Q.65 0.284 gm

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\4-TD\01-TC-TD-02 & 1 Eq.-(PQ)\Eng\4-Ionic Equilibrium\02_Ex.p65

EXERCISE # S-I

Q.1 0.209 M, 0.191 M, 9.13×10^{-3} M, 0

Q.2 pH = 11.48 , $[enH_2^{2+}] = 7.1 \times 10^{-8}$ M

Q.3 10.6

Q.4 $[S^{2-}] = 2.5 \times 10^{-15}$M

Q.5 2×10^{-4}M

Q.6 pH = 10.52

Q.7 $[OH^-] = 3.73 \times 10^{-2}$M, $[H_3PO_4] = 6 \times 10^{-18}$M

Q.8 8.35, 9.60, 4.66

Q.9 9.6

Q.10 $K_b \simeq 1.73 \times 10^{-5}$, 5.27

Q.11 $K_a \simeq 1.73 \times 10^{-5}$, 8.73

Q.12 10^{-5}M

Q.13. 10^{-3}M

Q.14 pH = 7.9 , pH = 7.3

Q.15 2.8×10^{-3} mole

Q.16 1.6×10^{-3}

Q.17 $K_d = 1/K_f = 4.8 \times 10^{-4}$

EXERCISE # O-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	B	C	C	B	C	D	D	C	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	D	B	B	C	C	B	C	C	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	D	D	C	C	C	B	B	A	D
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	B	C	C	C	B	C	D	A	C	C
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	B	C	D	A	B	D	B	C	A
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	C	A	A	A	B	D	A	A	B	C
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	D	A	D	C	D	D	D	C	A	C
Que.	71	72	73	74	75					
Ans.	B	B	D	B	B					

EXERCISE # O-II

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	C	D	B	C	A	A	B	D	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,B,C	A,B,C,D	B,C	A,B,D	D	A	A	B	B	B
Que.	21	22	23				24			
Ans.	C	A	A-R	B-P	C-Q	D-S	A-Q	B-P,S	C-R	D-T
Que.	25	26	27							
Ans.	D	B	C							

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\4-TD-01-TC-TD-02 & Eq - (PC)\Eng\4-Ionic Equilibrium\02_Ex.p65

EXERCISE # J-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	3	3	1	2	1	2	1	2	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	3	3	4	1	2	4	10.60	5.22 to 5.24	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	2	4	3	2	1	1	50	10	141	7
Que.	31	32								
Ans.	6021	108								

EXERCISE # J-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	Ph = 9	A	B	D	A	D	D	8	C,D
Que.	11	12	13	14	15	16	17	18	19	
Ans.	7	A	B	A	B	4.47	8.92 or 8.93	0.11	0.20	

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\4-TD-01-TC-TD-02 & JEE-(PC)\Eng\4-Ionic Equilibrium\02_Ex.p65